

Highlights

Discrete Learning Algorithms for Precipitation Estimation from Commercial Microwave Links

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- Rainfall reconstruction from CMLs is formulated as an explicit inverse optimization problem.
- A benchmark is built from EURADCLIM rain maps and synthetic link attenuations.
- The formulation preserves path geometry instead of collapsing links to point measurements.

Discrete Learning Algorithms for Precipitation Estimation from Commercial Microwave Links

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ABSTRACT

Commercial microwave links (CMLs) are part of the infrastructure of wireless networks. Their measured attenuations have been studied as an opportunistic source for monitoring spatiotemporal rainfall and other atmospheric phenomena. CML attenuation measurements can enhance the spatiotemporal accuracy and resolution of existing weather monitoring instruments. In addition, they serve as stand-alone weather monitoring devices in places where dedicated weather monitoring devices are scarce or do not exist.

Current techniques for 2D rainfall map reconstruction usually reduce CML measurements to virtual rain gauges, namely point measurements, and rely on interpolation techniques such as inverse distance weighting or Kriging. While effective in many scenarios, these methods are suboptimal because they do not address the modeling error introduced when a path-integrated attenuation measurement is replaced by a single-point rain-intensity value.

In this study, we revisit the rainfall map reconstruction problem from CML signal attenuation measurements with a principled optimization approach. We formulate the problem of partial-to-complete field reconstruction as a physics-informed optimization problem. The reconstructed rainfall field is quantized and represented by pixel rainfall variables whose values are constrained to agree with the observed CML signal attenuations. The resulting solution minimizes a weighted sum of attenuation errors along the links, spatial differences between neighboring pixels, and the total rainfall over the map.

To evaluate our approach, we create a benchmark of hundreds of rainfall maps and CML configurations. Rainfall maps are extracted from EURADCLIM by identifying rain events over patches of approximately $50 \times 50 \text{ km}^2$ across diverse terrain types and precipitation patterns. CMLs are overlaid using a publicly available Netherlands dataset, and attenuations are computed using the ITU-R P.838 model.

We then solve the inverse problem to reconstruct rainfall maps from attenuations and compare the results with interpolation baselines and an iteration-limited nonlinear adaptation of DCT-sparse reconstruction. Overall, this study reframes rainfall reconstruction from opportunistic sensing networks as a well-posed inverse problem with an explicit objective function. The framework also provides interpretability benefits compared to purely data-driven approaches.

1. Introduction

Accurate rainfall estimation is a major tool used for accurate hydrology studies, weather prediction, flood forecasting, and urban drainage management. Reliable information on the spatial and temporal structure of precipitation is particularly important in urban and densely populated environments, where short-duration convective events may produce strong local gradients and rapid hydrological response. Conventional weather monitoring tools, primarily rain gauges and weather radars, provide valuable information but also have well-known limitations. Rain gauges offer direct point measurements, and are considered very accurate when properly installed, but their spatial density is often insufficient for resolving localized rainfall variability. Weather radars provide wide-area coverage, yet their estimates are not at ground level, and are prone to many uncertainties such as scattering, blockages, unaccounted vertical variability of rainfall, and limited near-ground representativeness.

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Commercial microwave links (CMLs) offer a complementary source of rainfall information. These links are widely deployed as part of cellular telecommunication backhaul networks and continuously transmit electromagnetic signals between fixed antennas in a line-of-sight installments, nowadays at frequencies of 10 GHz to 90 GHz. During rainfall, raindrops affect the microwave radiation, causing measurable attenuation of the received signal. Since the frequencies range of operational links are sensitive to rain-induced attenuation ITU-R (2005); Gunn and East (1954); Olsen et al. (1978), the same infrastructure for communication was shown to also serve as an opportunistic environmental sensing network Overeem et al. (2013); Chwala and Kunstmann (2019); Fencil et al. (2015). As the CMLs are already deployed at large scale, often form dense networks in urban and suburban areas (especially in recent times, where dedicated CMLs networks for smart-city management are being deployed Ostrometzky and Messer (2024); Janco et al. (2022), and are gaining attention as part of the integrated sensing and communication (ISAC) paradigm Han et al. (2023)), they can provide near-ground measurements integrated along their paths, which span between tens of meters to several kilometers. Note that CMLs can be used in combination with satellite microwave links (SML) to further enhance their capabilities Nebuloni et al. (2025).

The physical basis for CML rainfall sensing is based on the attenuation of the signal that is caused by precipitation Gunn and East (1954); Olsen et al. (1978), and is described by the ITU-R power-law relation between rain rate and specific attenuation. Given the frequency and polarization of a link, the rain-induced attenuation accumulated along the link can be modeled as the path integral of the specific attenuation over the rainfall field. Thus, unlike a rain gauge, which measures rainfall at a point, a CML provides a path-integrated measurement. This distinction is fundamental. A single link does not directly observe the rainfall value at one location; rather, it constrains the average attenuation, and hence the rainfall intensity, along its propagation path. Consequently, rainfall reconstruction from CMLs is an inverse problem: the objective is to estimate a spatial rainfall field from a collection of line-integral-like measurements.

Many practical CML-based reconstruction methods simplify this inverse problem by first converting each link measurement into a single representative rain-rate value, often interpreted as a virtual rain gauge located at the link midpoint or at an assumed other point along the link path. Standard spatial interpolation techniques can then be applied to these virtual point observations. This approach is simple, computationally efficient, and often useful in practice. However, it also discards part of the geometric information contained in the link measurement. In particular, the fact that the attenuation is accumulated along an extended path is replaced by a point approximation. This simplification may introduce modeling errors, especially in heterogeneous rainfall fields or in regions where the geometry and density of the link network strongly influence the information available for reconstruction Gat and Messer (2019); Leijnse et al. (2008).

A more faithful treatment of CML data should account for the coupling between the unknown rainfall field and the measured attenuation along each link path. This coupling depends on the link geometry, length, frequency, polarization, and the spatial distribution of rain along the path. Reconstruction methods that explicitly incorporate this structure have the potential to use the available information more effectively, but they also lead to more challenging estimation problems. In particular, because the attenuation–rain-rate relation is nonlinear, the corresponding inverse problem may be formulated as a constrained nonconvex optimization problem over the rainfall field.

Evaluating such reconstruction methods is difficult in real operational settings because the true rainfall field is generally unknown. Real CML measurements also include additional effects that complicate the analysis, such as wet-antenna attenuation, baseline drift, quantization, hardware-dependent sampling, and measurement noise. While these effects are essential in operational systems, they make it difficult to isolate the intrinsic performance of the reconstruction algorithm itself. A controlled benchmark, in which the ground-truth rainfall field is known and the corresponding CML attenuations are generated through a forward physical model, provides a useful way to study this core inverse problem.

In this paper, we introduce such a benchmark framework. Starting from hourly EURADCLIM rainfall maps, we construct synthetic but physically grounded rain-event patches and generate corresponding noise-free CML attenuation measurements using the ITU-R forward model. This creates a controlled evaluation setting in which different reconstruction algorithms can be compared directly against the known rainfall field. By removing confounding measurement artifacts while preserving realistic spatial rainfall patterns, the proposed benchmark isolates the reconstruction problem and enables systematic analysis of the role of link geometry, interpolation assumptions, and optimization-based estimation.

We use this framework to compare standard inverse-distance weighting (IDW), a link-aware extension tailored to CML measurements, and a nonconvex optimization formulation that directly couples the estimated rainfall field to the

Table 1

ITU-R P.838-3 coefficients for horizontal polarization and resulting specific attenuation γ_R for selected rain rates (mm h^{-1}).

f (GHz)	κ_H	α_H	$\gamma_R(R = 1)$ (dB km $^{-1}$)	$\gamma_R(R = 2)$ (dB km $^{-1}$)	$\gamma_R(R = 4)$ (dB km $^{-1}$)	$\gamma_R(R = 8)$ (dB km $^{-1}$)
10	0.012	1.257	0.012	0.029	0.070	0.166
15	0.045	1.123	0.045	0.098	0.213	0.463
20	0.092	1.057	0.092	0.191	0.397	0.825
25	0.157	0.999	0.157	0.314	0.628	1.254
30	0.240	0.949	0.240	0.464	0.895	1.727
40	0.443	0.867	0.443	0.808	1.475	2.690

measured attenuations. In addition, we analyze reconstruction quality as a function of distance from the available links, thereby quantifying how the spatial arrangement of the CML network affects estimation accuracy.

1.1. Contributions

1. Introduction of a methodology for generating rain-event benchmarks for evaluation of rainfall estimation based on noise-free CML attenuation.
2. Creation of a benchmark (750 patches).
3. Introduction of a new algorithm ILDW - version of IDW tailored for CML's.
4. Introduction of rain-rate estimation as a nonconvex constrained optimization problem. Improve coupling between the estimation and measured attenuation.
5. Systematic evaluation of IDW, ILDW, and nonconvex optimization algorithms.
6. Evidence of quality of estimates as a function of distance from links.

2. Background and Related Work

2.1. ITU-R Forward Model: Rain-Induced Attenuation

During a rain event, the signal along a commercial microwave link experiences attenuation because raindrops absorb and scatter electromagnetic energy. We use Recommendation ITU-R P.838-3 for converting rain-rate into specific attenuation (ITU-R, 2005). The ITU-R model relates the rain rate R (in mm h^{-1}) to the *specific attenuation* γ_R (in dB km^{-1}) through the power law

$$\gamma_R = \kappa R^\alpha, \quad (1)$$

where the coefficients κ and α depend on the link frequency and polarization (ITU-R, 2005). See Table 1 for the coefficient values and resulting attenuation for horizontal polarization and rain rates of 1–8 mm h^{-1} .

The total rain-induced attenuation is the integral of γ_R along the link. Thus, longer links accumulate more attenuation, and links operating at different frequencies or polarizations respond differently to the same rainfall field. For a horizontal terrestrial link of length L with uniform rain rate R along the path, the total rain-induced attenuation is therefore $A = \gamma_R \cdot L$. If the rain-rate field is discretized into pixels, the total rain-induced attenuation is the sum of the attenuations over the segments obtained by intersecting the link with the pixels it traverses.

3. Benchmark Construction

In this section, we describe how we constructed our benchmark of rain-event patches combined with CML rain-attenuation. The resulting benchmark instances combine realistically derived rainfall maps with real-world link configurations, from which noise-free rain-induced attenuation measurements are generated for every link.

3.1. Data Source

We extract rain events from the European climatological gauge-adjusted radar precipitation dataset EURADCLIM (Overeem et al., 2023). EURADCLIM is derived from OPERA European radar composites and provides hourly precipitation fields at 2 km spatial resolution over a large fraction of Europe. The dataset are distributed in HDF5 format according to the OPERA Data Information Model (ODIM) convention (EUMETNET OPERA, 2026). File-level access and documentation are provided through the KNMI data platform (KNMI, 2023).

Patch size (km)	# links	1-multiplicity	2-multiplicity
50×50	939	99	420
62×60	1254	128	563
72×70	1507	157	675

Table 2

Representative link multiplicities for three patch dimensions. The multiplicity of a link is defined as the number of frequencies assigned to it. The second column reports the total number of link–frequency pairs. In the data set, link multiplicities were either 1 or 2, with approximately 10% of links having multiplicity 1.

3.2. Automatic Extraction of Rain Events

To obtain benchmark instances of manageable and comparable size, hourly maps are processed independently and rain events are extracted as spatial patches. We threshold a locally averaged rainfall field, compute connected components of the thresholded pixels, and keep only components whose bounding-box side lengths fall within a prescribed size. In this way, each retained event corresponds to a rain patch.

3.3. Refinement and Smoothing

We refine and smooth each rain patch derived from EURADCLIM rainfall maps by subdividing each pixel into 16×16 subpixels (each $125 \times 125 \text{ m}^2$), followed by Gaussian smoothing with $\sigma = 1$ in subpixel units. The resulting rain patches are treated as the ground-truth rainfall field.

3.4. CML Dataset

CML metadata (including link locations, lengths, and frequencies) are obtained from the publicly available 4TU dataset (Overeem et al., 2016) (we assumed horizontal polarization for all links). For each rainfall patch, the set of links is cropped to the patch extent and overlaid on the rainfall field. The ITU-R rain-attenuation model is then applied along each link path to compute the corresponding attenuation values.

4. Benchmark Statistics

In this section, we summarize key properties of the benchmark patches and their associated link geometry. We describe the geographic distribution and size range of the selected rain events, illustrate representative patch instances, summarize link statistics and topology, and report how rainy and dry pixels are distributed with respect to distance from the third-closest link.

Patches.

Time. The rain patches were extracted from hourly EURADCLIM maps between 18 and 31 January 2023.

Geographic distribution. See Figure 1 for a depiction of the geographic locations of the 100 rain patches.

Patch sizes. A scatter diagram of patch side lengths is depicted in Figure 2. The smallest patch is $50 \times 50 \text{ km}$ and the largest is $88 \times 88 \text{ km}$.

Sample patches. Sample rain-events from the benchmark are depicted in Figures 11 and 13 together with their geographic location.

Links. A depiction of the links in the largest $88 \times 88 \text{ km}$ patch appears in Figure 3. The pixels in this figure are binned according to their distance to the third-closest link. For a pixel p , let $d_3(p)$ denote the distance from the pixel center to the third-closest link. Equivalently, $d_3(p)$ is the smallest radius such that a disk centered at p intersects at least three links.

Table 3 provides data about the distribution of link lengths, frequencies and endpoint degrees in this patch (the degree of an endpoint is number of links incident to this endpoint). See Table 2 for the number of links and their multiplicities for typical patch sizes (multiplicity 2 means a link operating in two frequencies).

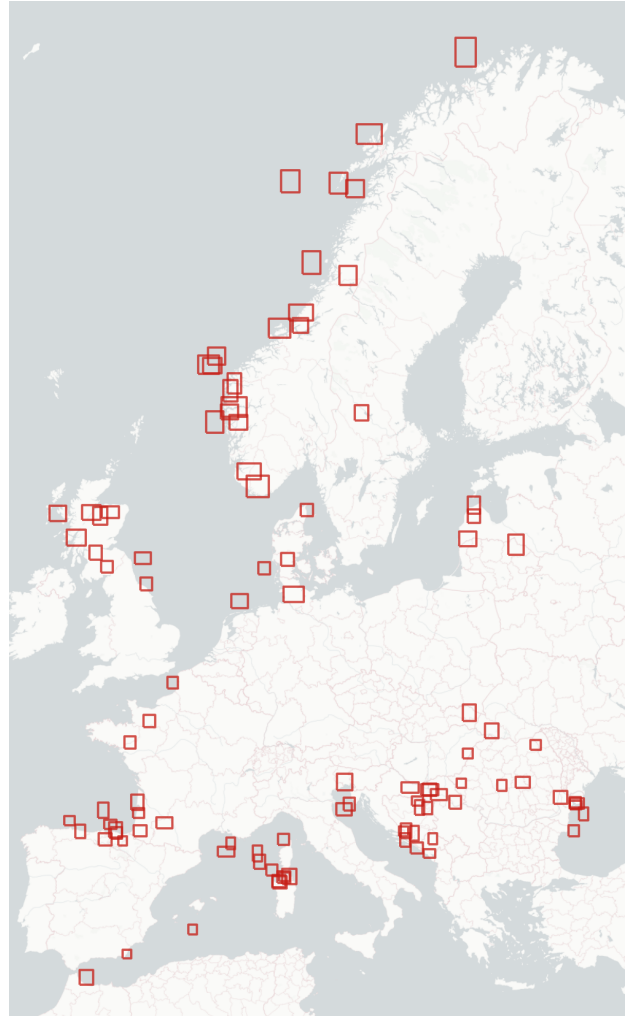


Figure 1: Geographic locations of the 100 benchmark patches.

Length bin (km)	Count	%
[0, 1)	330	17.4
[1, 2)	515	27.2
[2, 4)	546	28.8
[4, 8)	387	20.4
[8, 16)	109	5.8
[16, 24]	7	0.4

Frequency bin (GHz)	Count	%
[10, 15)	20	1.1
[15, 20)	67	3.5
[20, 25)	79	4.2
[25, 30)	257	13.6
[30, 35)	84	4.4
[35, 40)	1306	69.0
[40, 60]	81	4.3

Endpoint-degree bin	Count	%
1	168	15.0
2	560	49.9
3, 4	246	21.9
5, 6	68	6.1
7..10	39	3.5
11..20	26	2.3
≥ 21	16	1.4

Table 3

Link statistics for the largest-patch. The first panel counts links by path-length interval, the second counts links by carrier-frequency interval, and the third counts degrees of link endpoints.

Rainy/Dry Pixel Distance Stratification. We considered all 100 benchmark patches and classified the pixels in each patch as rainy or dry using a rainfall threshold of 0.6 mm h^{-1} . Each of these two pixel sets was then partitioned into bins according to the pixel-level distance $d_3(p)$. The mean and standard deviation of the rainy-pixel and dry-pixel counts in each distance bin, computed across all patches, are reported in Table 4.

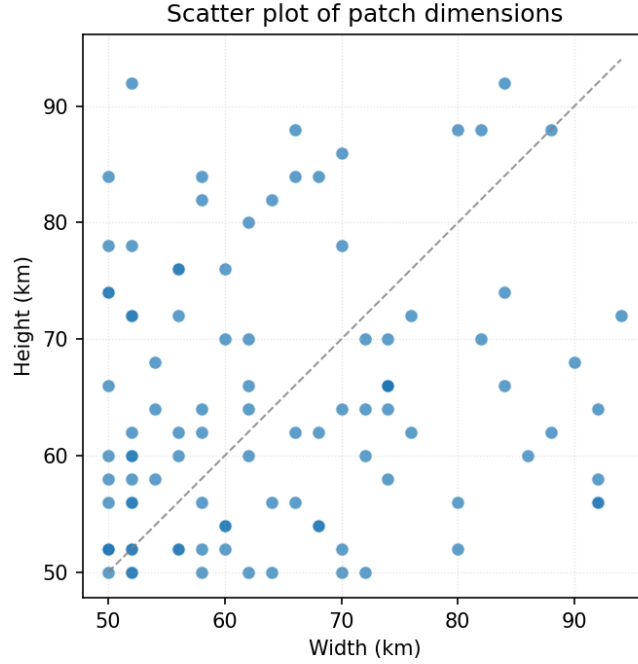


Figure 2: Patch width versus patch height for the 100 benchmark patches, with one dot per patch. Across patches, the mean patch area is 269,491 pixels with SD of 76,755 pixels (pixels are squares with side length 125 m).

d_3 bin (m)	Rainy pixels			Non-rainy pixels		
	Mean count	Mean %	SD	Mean count	Mean %	SD
[0, 125]	4267	1.790	960	564	1.840	778
(125, 375]	16 738	7.010	3803	2261	7.400	3094
(375, 750]	26 810	11.220	6707	3389	11.090	4710
(750, 1500]	51 989	21.760	14 468	6447	21.090	9258
(1500, 3125]	88 839	37.180	27 531	10 776	35.250	15 554
(3125, 6000]	45 929	19.220	15 679	6197	20.270	8990
(6000, 9000]	3836	1.610	2726	850	2.780	1290
(9000, ∞)	514	0.220	967	86	0.280	251
Total	238 922	100.000		30 569	100.000	

Table 4

Ground-truth rainy and non-rainy pixel counts per d_3 bin. Values are mean and standard deviation across patches.

5. Inverse Solvers: From Link Attenuations to Rainfall Maps

In this section, we present three solvers for the inverse problem: IDW, ILDW, and a nonconvex optimization model.

5.1. Problem Formulation

The inputs for the inverse problem consists of a set of links $\mathcal{L}$ in a patch P (set of pixels). Every link ℓ has an attenuation A_ℓ and the following attributes: frequency, polarization, endpoints and length $|\ell|$.

The output is an estimated rain-rate field $\hat{R} : P \rightarrow \mathbb{R}_{\geq 0}$.

5.2. Rain Equivalent for Link Attenuation

The *rain-rate equivalent* R_ℓ of link ℓ is the uniform rain-rate along the link that would cause the link total attenuation A_ℓ . This rain-rate is derived from the ITU-R model by:

$$R_\ell \triangleq \left(\frac{A_\ell / |\ell|}{\kappa_\ell} \right)^{1/\alpha_\ell},$$

where $(\kappa_\ell, \alpha_\ell)$ the ITU-R P.838-3 coefficients corresponding to the link's frequency and polarization.

5.3. IDW

Inverse-distance weighting (IDW) is one of the standard interpolation strategies used in commercial-microwave-link rainfall reconstruction (Zimmerman et al., 1999; Guo-Dong et al., 2003; Chen and Liu, 2012; Chwala and Kunstmann, 2019; Eshel et al., 2020, 2021). The input consists of a set of links, where each link ℓ carries a total attenuation, frequency, and polarization.

IDW for CML's replaces every link ℓ with a virtual rain-gauge g_ℓ positioned at the midpoint between the endpoints of ℓ . The rain-rate associated with g_ℓ is simply the rain-rate equivalent R_ℓ . Inverse-distance weighting is applied with the virtual gauges positions and rain-rates as inputs.

We consider a setting in which the weight of a gauge decreases quadratically (i.e., decay exponent equals 2) and the maximum radius of influence $r_{\max}$ is set to 15 km. Distances to pixels are measured to the center of the pixel. If one or more link midpoints lie inside pixel p , then the estimated rainfall in pixel p is set to the mean of their rainfall estimates. The pseudocode for IDW appears in Algorithm 1.

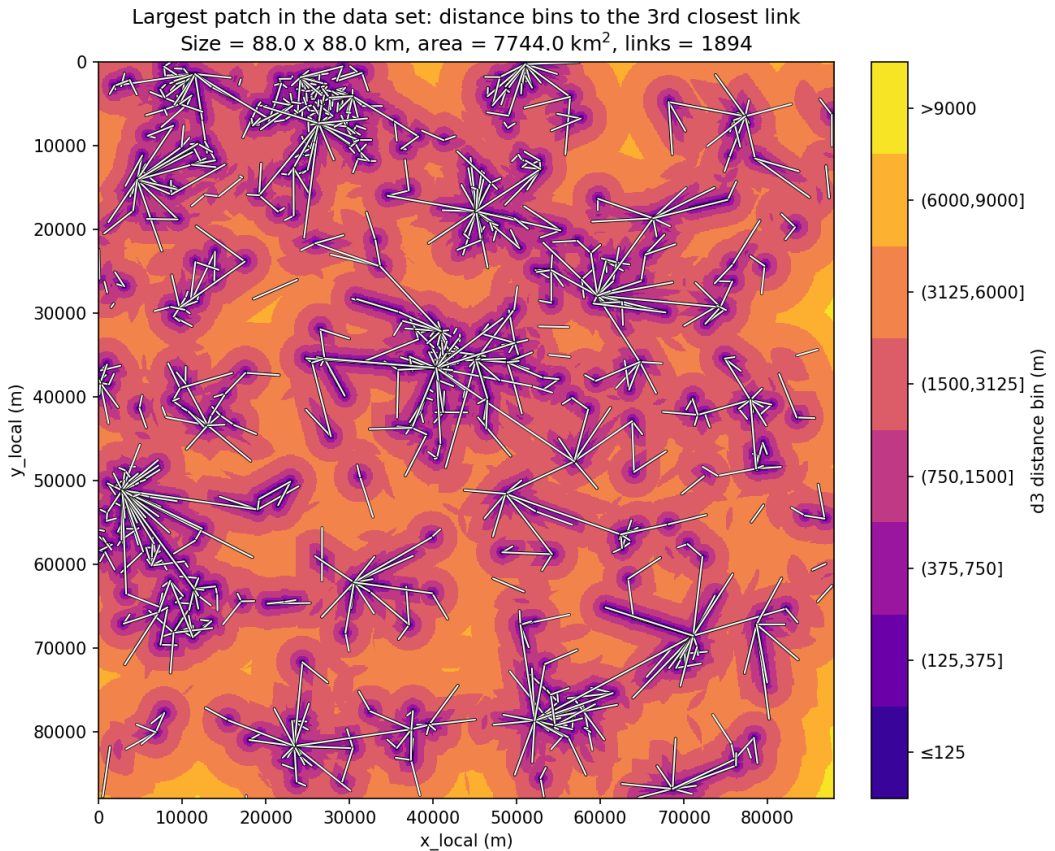


Figure 3: d_3 -bin map for the largest evaluated patch in the benchmark. Each pixel is colored according to its d_3 bin, and the complete link geometry is overlaid in white.

Algorithm 1: Discretized IDW. Each link is replaced with a virtual gauge located at the midpoint of the link. The rain-rate at the gauge is set to the rain-rate equivalent of its link. IDW is then applied with respect to these virtual gauges. We set the exponent of the decay to 2.

Data: link set $\mathcal{L}$, a patch P , and a maximum influence distance $r_{\max}$

Result: rainfall field $\hat{R} : P \rightarrow \mathbb{R}_{\geq 0}$

```

1 foreach link  $\ell \in \mathcal{L}$  do                                     // compute rain-rate equivalent per link
2    $(\ell.\kappa, \ell.\alpha) \leftarrow \text{ITU838}(\ell.\text{freq}, \ell.\text{polarization})$ 
3    $R_\ell \leftarrow \left( \frac{A_\ell / |\ell|}{\ell.\kappa} \right)^{1/\ell.\alpha}$ 
4 foreach pixel  $p \in P$  do                                     // compute rain  $\hat{R}(p)$ 
5    $M_p \leftarrow \{\ell \in \mathcal{L} : \ell.\text{midpoint} \in p\}$ 
6   if  $M_p \neq \emptyset$  then
7      $\hat{R}(p) \leftarrow \frac{1}{|M_p|} \sum_{\ell \in M_p} R_\ell$                      // pixel contains a link endpoint
8   else
9      $N_p \leftarrow \{\ell \in \mathcal{L} : \|\ell.\text{midpoint} - p.\text{center}\|_2 \leq r_{\max}\}$ 
10    if  $N_p = \emptyset$  then
11       $\hat{R}(p) \leftarrow 0$                                          // pixel far from links
12    else
13       $\hat{R}(p) \leftarrow \sum_{\ell \in N_p} \frac{\|\ell.\text{midpoint} - p.\text{center}\|_2^{-2}}{\sum_{\ell' \in N_p} \|\ell'.\text{midpoint} - p.\text{center}\|_2^{-2}} \cdot R_\ell$ 

```

5.3.1. ILDW

We propose a variation of IDW that alleviates the inherent mismatch between the measured and IDW-estimated link attenuation (see Appendix A.1). We refer to this variation as inverse link-distance weighting (ILDW). ILDW assigns a weight to every link that is inversely proportional to the distance between the pixel and the link, where

$$\text{dist}(p, \ell) \triangleq \min_{q \in \ell} \|p.\text{center} - q\|.$$

As in IDW, we set the decay exponent to 2 and the maximum radius of influence $r_{\max}$ is set to 15 km. If one or more links cross the pixel, the estimated rain-rate is taken to be the mean of the corresponding link rain-rate equivalents. Otherwise, the estimate is a weighted average over links within distance $r_{\max}$ (set to 15km). The pseudocode for ILDW is shown in Algorithm 2, with lines that differ from IDW highlighted in red.

Algorithm 2: Discretized ILDW. Rainfall is averaged by inverse squared distance to a link. Lines that are different from their corresponding lines in IDW are colored red.

Data: link set $\mathcal{L}$, a patch P , and a maximum influence distance $r_{\max}$

Result: rainfall field $\hat{R} : P \rightarrow \mathbb{R}_{\geq 0}$

```

1 foreach link  $\ell \in \mathcal{L}$  do                                     // compute rain-rate equivalent per link
2    $(\ell.\kappa, \ell.\alpha) \leftarrow \text{ITU838}(\ell.\text{freq}, \ell.\text{polarization})$ 
3    $R_\ell \leftarrow \left( \frac{A_\ell / |\ell|}{\ell.\kappa} \right)^{1/\ell.\alpha}$ 
4 foreach pixel  $p \in P$  do                                     // compute rain  $\hat{R}(p)$ 
5    $X_p \leftarrow \{\ell \in \mathcal{L} : \ell \cap p \neq \emptyset\}$ 
6   if  $X_p \neq \emptyset$  then
7      $\hat{R}(p) \leftarrow \frac{1}{|X_p|} \sum_{\ell \in X_p} R_\ell$                      // at least one link traverses the pixel
8   else
9      $N_p \leftarrow \{\ell \in \mathcal{L} : \text{dist}(p, \ell) \leq r_{\max}\}$ 
10    if  $N_p = \emptyset$  then
11       $\hat{R}(p) \leftarrow 0$                                          // pixel far from links
12    else
13       $\hat{R}(p) \leftarrow \sum_{\ell \in N_p} \frac{\text{dist}^{-2}(p, \ell)}{\sum_{\ell' \in N_p} \text{dist}^{-2}(p, \ell')} \cdot R_\ell$ 

```

5.3.2. Nonconvex Formulation

We formulate the inverse problem as a constrained optimization problem. The decision variables are the estimated rain-rates $\hat{R}(p)$ for all pixels $p \in P$. The constraints are simply nonnegativity constraints that reflect the physical requirement that rainfall is nonnegative. The objective is to minimize $J(\hat{R})$, where the cost function J consists of the 4 weighted terms:

$$J(\hat{R}) = w_{\text{atten}} \cdot J_{\text{atten}}(\hat{R}) + w_{1d} \cdot J_{1d}(\hat{R}) + w_{2d} \cdot J_{2d}(\hat{R}) + w_{\text{total}} \cdot J_{\text{total}}(\hat{R}).$$

The terms in $J(\hat{R})$ are as follows:

1. The attenuation term J_{atten} quantifies disagreement between observed and estimated link attenuations. For a rain-rate field estimate $\hat{R} : P \rightarrow \mathbb{R}_{\geq 0}$ and a link ℓ , let $A_{\ell}(\hat{R})$ denote the attenuation induced on ℓ by $\hat{R}$. This attenuation is computed using the ITU-R forward model with the link's frequency and polarization. The attenuation term J_{atten} is defined by

$$J_{\text{atten}}(\hat{R}) \triangleq \frac{1}{|\mathcal{L}|} \sum_{\ell \in \mathcal{L}} \frac{(A_{\ell}(\hat{R}) - A_{\ell})^2}{|A_{\ell}|}.$$

2. The first-difference term J_{1d} enforces spatial smoothness, and is defined by

$$J_{1d}(\hat{R}) \triangleq \frac{1}{|P|} \sum_{(u,v) \in \mathcal{E}(P)} (\hat{R}(u) - \hat{R}(v))^2,$$

where $\mathcal{E}(P)$ is the set of horizontally and vertically adjacent pixel pairs in patch P .

3. The second-difference term J_{2d} penalizes curvature, and is defined by

$$J_{2d}(\hat{R}) = \frac{1}{|P|} \sum_{(p,q,t) \in \mathcal{T}(P)} ((\hat{R}(q) - \hat{R}(p)) - (\hat{R}(t) - \hat{R}(q)))^2,$$

where $\mathcal{T}(P)$ is the set of horizontal and vertical triples of collinear pixels in patch P .

4. The shrinkage term J_{total} biases the solution toward smaller overall rainfall values, and is defined by

$$J_{\text{total}}(\hat{R}) \triangleq \frac{1}{|P|} \sum_{p \in P} \hat{R}(p)^2.$$

Implementation Details:

1. The weight of each term is set to the reciprocal of its value evaluated at the ILDW solution. We further divide w_{2d} by 2 and w_{total} by $|P|$. This normalizes the objective instance-by-instance.
2. We use the L-BFGS-B algorithm (Byrd et al., 1995; Zhu et al., 1997), a limited-memory quasi-Newton method for smooth constrained optimization, as implemented in SciPy (Virtanen et al., 2020).

Nonconvex Cost Function: The ITU-R forward model (Equation (1)) attaches an attenuation exponent α to every link based on its frequency and polarization. When $\alpha = 1$, the predicted attenuation is linear in the rain-rate variables, so J_{atten} is a sum of squares of linear residuals and is convex. For $\alpha \neq 1$, the power-law attenuation makes the residual nonlinear, and the attenuation term is generally nonconvex. The nonconvexity emphasizes the importance of the initial solution used by the solver because local minima need not be global minima.

We illustrate this nonconvexity with a concrete 3×3 single-link example in Appendix A.2.

Summary of the Nonconvex Inverse Solver. The input to the inverse solver is a patch P and a set $\mathcal{L}$ of links. We treat each link $\ell \in \mathcal{L}$ as a record containing its geometry and physical attributes that include its frequency f_{ℓ} , polarization, length $|\ell|$, and measured attenuation A_{ℓ} . The L-BFGS-B solver also takes an initial rain-rate field $\hat{R}_{\text{init}}$ as input. We write

$$\hat{R} \leftarrow \text{LBFGSB}(\mathcal{L}, P, \hat{R}_{\text{init}})$$

to denote one invocation of the solver and the resulting rain-rate field estimate.

Coping with nonconvexity. We use three strategies to deal with the nonconvexity of J_{atten} :

1. Use ILDW as the initial solution. We refer to this solver as SOLVER(ILDW).
2. We initialize the nonconvex solver with the ground-truth rainfall field. This configuration, denoted SOLVER(GT), serves as a best-case reference for the nonconvex formulation.
3. Obtain a convex surrogate objective by replacing each link with a virtual link that keeps the same geometry but changes its frequency to f^0 , chosen so that the attenuation exponent equals 1 (i.e., $f^0 = 25$ GHz for horizontal polarization). We refer to this solver as CONVEX SOLVER.
4. Apply homotopy continuation to smoothly transition from the convex surrogate to the original nonconvex formulation. We refer to this solver as HOMOTOPY SOLVER.

We elaborate on the last two strategies below.

5.4. CONVEX SOLVER: Convex Surrogate Inverse Solver

We construct a virtual link set $\mathcal{L}^0$ from the original link set $\mathcal{L}$. For each link $\ell \in \mathcal{L}$, the corresponding virtual link $\ell^0 \in \mathcal{L}^0$ keeps the same geometry and length, but uses the frequency f^0 for which the ITU-R attenuation exponent equals 1.¹ The attenuation of the virtual link ℓ^0 is set to

$$A_{\ell}^0 \triangleq \kappa_0 \cdot R_{\ell} \cdot |\ell|,$$

where κ_0 is the ITU-R coefficient at frequency f^0 and R_{ℓ} is the rain-rate equivalent of the original link ℓ (see Section 5.2). For $\mathcal{L}^0$, the attenuation error term J_{atten} is a sum of squares and is therefore convex. We write

$$\hat{R}^0 \leftarrow \text{LBFGSB}(\mathcal{L}^0, P, \hat{R}_{\text{init}})$$

to denote applying L-BFGS-B to the convex surrogate inverse problem. The surrogate objective is convex and includes the strictly convex J_{total} , so it has a single local minimum, which is also the global minimum. Thus, the initial solution $\hat{R}_{\text{init}}$ affects only the convergence speed of L-BFGS-B (In our implementation, $\hat{R}_{\text{init}} = 0.6$ for all pixels.). We refer to the resulting solver as CONVEX SOLVER.

5.5. HOMOTOPY SOLVER: Smooth Transition from Convex to Nonconvex

We extend the virtual-link construction to a parameter $\lambda \in [0, 1]$. For each λ , let

$$\mathcal{L}^{\lambda} \triangleq \{\ell^{\lambda} : \ell \in \mathcal{L}\}.$$

The virtual link ℓ^{λ} keeps the geometry, length, and polarization of ℓ , but has frequency

$$f_{\ell}^{\lambda} \triangleq (1 - \lambda)f^0 + \lambda f_{\ell}.$$

Let $(\kappa_{\ell}(\lambda), \alpha_{\ell}(\lambda))$ be the ITU-R coefficients at frequency f_{ℓ}^{λ} . The attenuation of ℓ^{λ} is set to

$$A_{\ell}^{\lambda} \triangleq |\ell| \cdot \kappa_{\ell}(\lambda) \cdot R_{\ell}^{\alpha_{\ell}(\lambda)}.$$

Algorithm 3: HOMOTOPY SOLVER- a solver that applies a smooth transition from convex optimization to optimizing the nonconvex objective. Assume the same polarization for all the links.

Data: link set $\mathcal{L}$, patch P , initial field $\hat{R}_{\text{init}}$, number of homotopy steps n

Result: rainfall field $\hat{R} : P \rightarrow \mathbb{R}_{\geq 0}$

```

1  $\hat{R}^0 \leftarrow \text{LBFGSB}(\mathcal{L}^0, P, \hat{R}_{\text{init}})$ 
2 for  $i = 1, \dots, n$  do
3    $\hat{R}^{i/n} \leftarrow \text{LBFGSB}(\mathcal{L}^{i/n}, P, \hat{R}^{(i-1)/n})$ 
4 return  $\hat{R}^1$ 
```

¹Note that f^0 depends on the link's polarization. For simplicity, we consider a single polarization.

5.6. DCT-sparse baseline adapted from Roy et al.

We additionally evaluate a no-dynamics sparsity baseline motivated by Roy et al. (2016). Their Eq. (26) uses a linear measurement operator, an orthonormal DCT basis and nonnegative reconstructed rainfall. Here the measurement term is adapted to the existing nonlinear ITU observations:

$$\min_{u \geq 0} \frac{1}{\sigma_e^2} \|A - F(u)\|_2^2 + \lambda \|Du\|_1, \quad F_\ell(u) = k_\ell \sum_j l_{\ell j} u_j^{\alpha_\ell}.$$

We use an orthonormal separable two-dimensional DCT-II, including the DC coefficient, $\lambda = 2$ and $\sigma_e^2 = 10^{-5}$. This retains the prior structure but is not a reproduction of the paper's linear experiment. The DCT layout is an explicit implementation choice. No temporal information, learned prior, pixel-count normalization or length-normalized data residual is used. Fast transforms and sparse operators avoid constructing the dense basis. Successive linearization yields convex ADMM/CG subproblems; damping and backtracking control the nonlinear updates. The method starts from a constant 0.6 mm/h field. Settings are transferred rather than tuned to this benchmark. All 100 saved runs stopped at the outer-iteration limit (50 steps), with `success=false`. We therefore label this method "Roy DCT (limited)" and retain all runs rather than selecting favorable cases. These results assess the saved nonlinear implementation and iteration budget, not the attainable accuracy of a converged Roy estimator.

6. Experimental Results and Analysis

We compare seven saved reconstruction methods on the same 100 benchmark patches: IDW, ILDW, Solver(ILDW), Solver(GT), the convex and homotopy solvers, and the iteration-limited nonlinear Roy DCT adaptation. Ground truth is the rainfall field used to generate the synthetic link attenuations. Each patch receives equal weight in the aggregate metrics. Solver(GT) is a privileged initialization reference, not an operational baseline.

6.1. Notation

For a patch P and solver s , let $R_P(p)$ denote the ground-truth for pixel p . Let $\hat{R}_{P,s}(p)$ denote the estimate returned by solver s for the pixel $p \in P$. Define the means:

$$\overline{\hat{R}}_{P,s} \triangleq \frac{1}{|P|} \sum_{p \in P} \hat{R}_{P,s}(p), \quad \overline{R}_P \triangleq \frac{1}{|P|} \sum_{p \in P} R_P(p).$$

For a patch P and solver s , we compute three values: the root mean squared error

$$\text{RMSE}_{P,s} \triangleq \sqrt{\frac{1}{|P|} \sum_{p \in P} (\hat{R}_{P,s}(p) - R_P(p))^2},$$

the signed bias

$$\text{Bias}_{P,s} \triangleq \frac{1}{|P|} \sum_{p \in P} (\hat{R}_{P,s}(p) - R_P(p)),$$

and the Pearson correlation coefficient

$$r_{P,s} \triangleq \frac{\sum_{p \in P} (\hat{R}_{P,s}(p) - \overline{\hat{R}}_{P,s})(R_P(p) - \overline{R}_P)}{\sqrt{\sum_{p \in P} (\hat{R}_{P,s}(p) - \overline{\hat{R}}_{P,s})^2} \sqrt{\sum_{p \in P} (R_P(p) - \overline{R}_P)^2}}.$$

For a pixel p , the absolute error (AE) is defined by

$$AE_{P,s}(p) \triangleq |\hat{R}_{P,s}(p) - R_P(p)|,$$

and the relative absolute error (RAE) is defined by

$$RAE_{P,s}(p) \triangleq \frac{AE_{P,s}(p)}{R_P(p)}.$$

To avoid division by zero, we use the absolute error for nonrainy pixels.

Method	RMSE	Signed bias	Pearson r
IDW	1.307 ± 1.241	0.034 ± 0.298	0.866 ± 0.060
ILDW	1.239 ± 1.188	0.047 ± 0.263	0.878 ± 0.055
Solver(ILDW)	1.059 ± 1.198	0.040 ± 0.236	0.918 ± 0.056
Solver(GT)	0.503 ± 0.371	0.009 ± 0.030	0.961 ± 0.032
Convex solver	0.845 ± 0.807	0.023 ± 0.075	0.940 ± 0.035
Homotopy solver	0.845 ± 0.807	0.023 ± 0.075	0.940 ± 0.035
Roy DCT (limited)	3.554 ± 1.865	-2.443 ± 0.786	0.165 ± 0.060

Table 5

Whole-patch metrics: mean $\pm$ sample standard deviation across 100 patches, with equal patch weights. RMSE and bias are in mm/h. Roy DCT denotes the nonlinear adaptation at its iteration limit; Solver(GT) uses privileged ground-truth initialization.

Method vs. IDW	Wins:losses	$\bar{\Delta}$	p (one-sided)	99% upper bound
ILDW	97:3	-0.068	2.4e-12	-0.047
Solver(ILDW)	98:2	-0.248	6e-21	-0.198
Solver(GT)	100:0	-0.804	2.2e-11	-0.547
Convex solver	100:0	-0.461	1.9e-15	-0.344
Homotopy solver	100:0	-0.461	1.9e-15	-0.344
Roy DCT (limited)	0:100	2.247	1	2.438

Table 6

Paired patchwise RMSE comparison, with $\Delta = \text{RMSE}_s - \text{RMSE}_{\text{ILDW}}$. Negative differences favor the method. Tests use the alternative $E[\Delta] < 0$; values are unadjusted for multiple comparisons. Iteration-limited Roy outputs are retained.

Method	FP/all (%)	FN/all (%)	FP/dry (%)	FN/wet (%)
IDW	3.96	0.24	59.83	0.32
ILDW	3.70	0.29	56.24	0.38
Solver(ILDW)	3.24	0.31	46.30	0.41
Solver(GT)	1.14	0.21	33.05	0.26
Convex solver	2.28	0.38	40.02	0.50
Homotopy solver	2.28	0.38	40.07	0.50
Roy DCT (limited)	4.85	34.91	48.16	39.12

Table 7

Mean patchwise wet/dry error rates at 0.6 mm/h. False positive: predicted wet, observed dry; false negative: predicted dry, observed wet. Conditional rates use their respective ground-truth classes.

6.2. Whole-patch accuracy

The saved Roy DCT outputs have mean RMSE 3.554 mm/h, mean signed bias -2.443 mm/h, and mean Pearson correlation 0.165. Their strong negative bias and low correlation show that close attenuation agreement alone does not determine a useful full rainfall map. The comparison is provisional because none of these runs declared convergence. Among the operational methods in this cache, the convex and homotopy solvers have the smallest mean whole-patch RMSE (approximately 0.845 mm/h), compared with 1.307 mm/h for IDW.

6.3. Paired comparison with IDW

These tests describe the saved outcomes; they do not remove the numerical limitations of the Roy runs or establish a ranking of fully converged methods.

6.4. Wet/dry classification and attenuation fit

The Roy adaptation achieves small attenuation residuals despite poorer map accuracy. Under sparse link coverage, fitting path integrals is insufficient to determine rainfall between links. Its DCT penalty, initialization, transferred

Method	Attenuation MAE (dB/km)	J_{atten} (dB ² /km)
IDW	0.0503	0.0528
ILDW	0.0136	0.0042
Solver(ILDW)	0.00207	0.000295
Solver(GT)	0.00129	4.95e-05
Convex solver	0.00107	3.35e-05
Homotopy solver	0.000747	1.33e-05
Roy DCT (limited)	0.000388	9.06e-07

Table 8

Mean patchwise attenuation errors, evaluated consistently for all methods using the actual ITU forward model. $J_{\text{atten}} = M^{-1} \sum_{\ell} (\hat{A}_{\ell} - A_{\ell})^2 / L_{\ell}$. This comparison metric is not the full native objective of Roy DCT.

hyperparameters and finite iteration budget all affect the result. These experiments do not isolate which of those choices accounts for the observed underestimation.

6.5. Distance-conditioned errors

Pixels are grouped by distance d_3 to the third-closest link. Rainy pixels have ground-truth rainfall at least 0.6 mm/h; for these pixels we use relative absolute error. For nonrainy pixels we use absolute error. Each patch contributes one median per nonempty distance bin. The figures below use the same seven-method cache and binning rules as the numerical summaries.

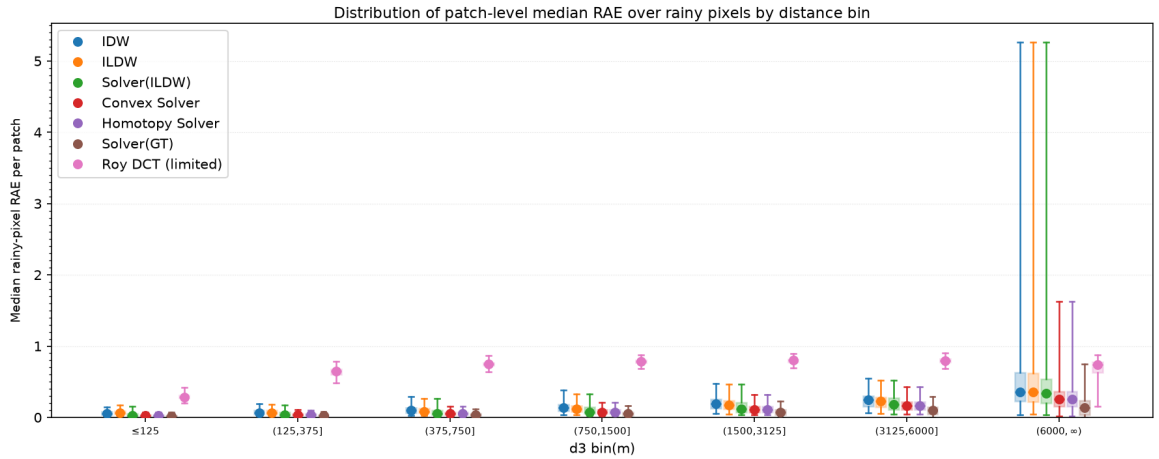


Figure 4: Patchwise median relative absolute errors for rainy pixels, stratified by d_3 . Roy DCT denotes iteration-limited nonlinear outputs.

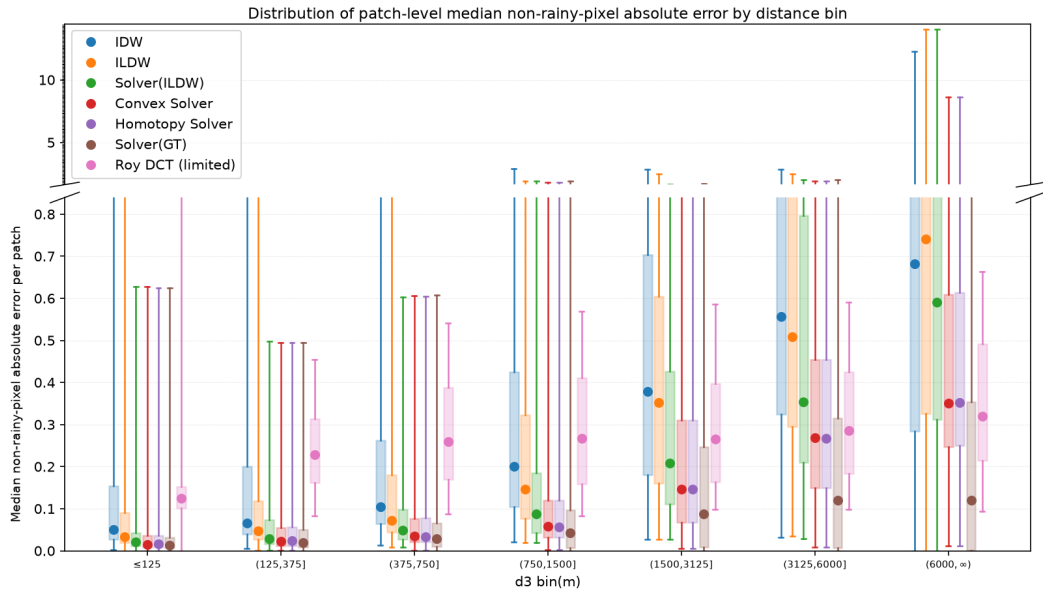
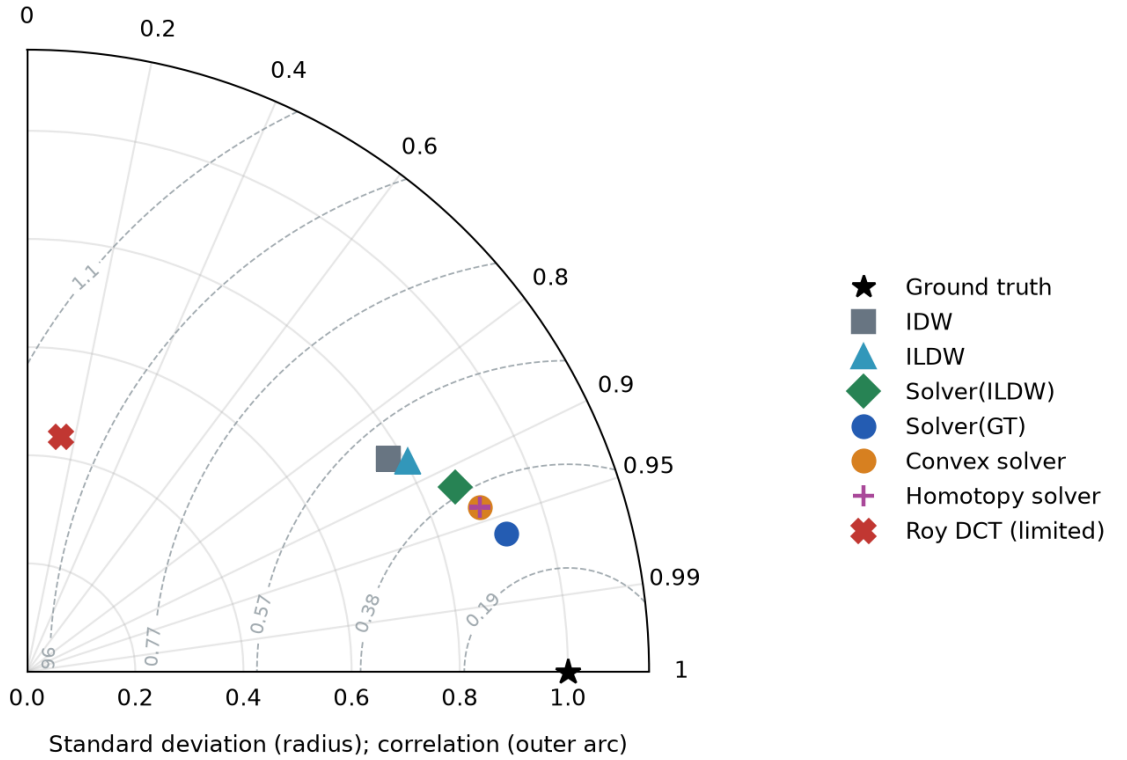


Figure 5: Patchwise median absolute errors for nonrainy pixels, stratified by d_3 . Empty strata do not contribute a patch value.

6.6. Taylor diagrams

A Taylor diagram summarizes within-patch standard deviation, correlation and centered RMSE. Let g_i and f_i be the ground-truth and reconstructed standard deviations for patch i , and c_i their covariance, using population moments over pixels. With equal patch weights, the combined unnormalized point uses $G = \sqrt{\langle g_i^2 \rangle}$, $F = \sqrt{\langle f_i^2 \rangle}$ and $C = \langle c_i \rangle$, with angle $\arccos(C/(GF))$ and radius F . Its centered error is $E_c = \sqrt{F^2 + G^2 - 2C}$, the root mean square of the patchwise centered errors. It is not the arithmetic mean of patch correlations. For the normalized diagram, each patch is first divided by its own ground-truth standard deviation, yielding $G = 1$, $F = \sqrt{\langle f_i^2 / g_i^2 \rangle}$ and $C = \langle c_i / g_i^2 \rangle$. Neither diagram encodes mean bias; for every patch, $\text{RMSE}_i^2 = E_{c,i}^2 + \text{Bias}_i^2$.

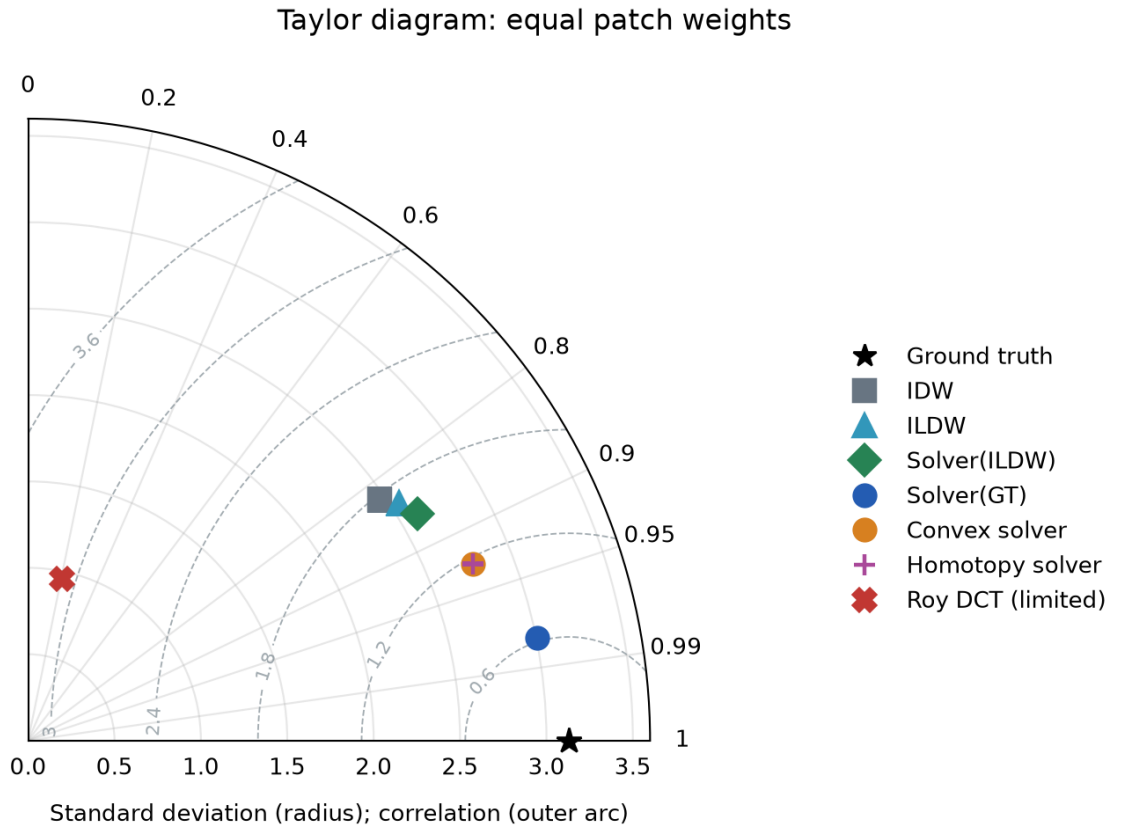
Taylor diagram: equal patch weights



Normalized by each patch's ground-truth SD

Dashed contours: centered RMSE. Bias is excluded. Roy: all 100 runs iteration-limited.

Figure 6: Combined normalized Taylor diagram, with equal patch weights. Dashed contours indicate centered RMSE and the star denotes ground truth. Bias must be read separately from Table 5.



Unnormalized: standard deviations and centered RMSE in mm/h

Dashed contours: centered RMSE. Bias is excluded. Roy: all 100 runs iteration-limited.

Figure 7: Combined unnormalized Taylor diagram in mm/h. Equal patch weights are retained, but patches with larger physical variance contribute more to the variance moments. The convex and homotopy points nearly coincide.

Normalized Taylor diagrams: 100 patches per method

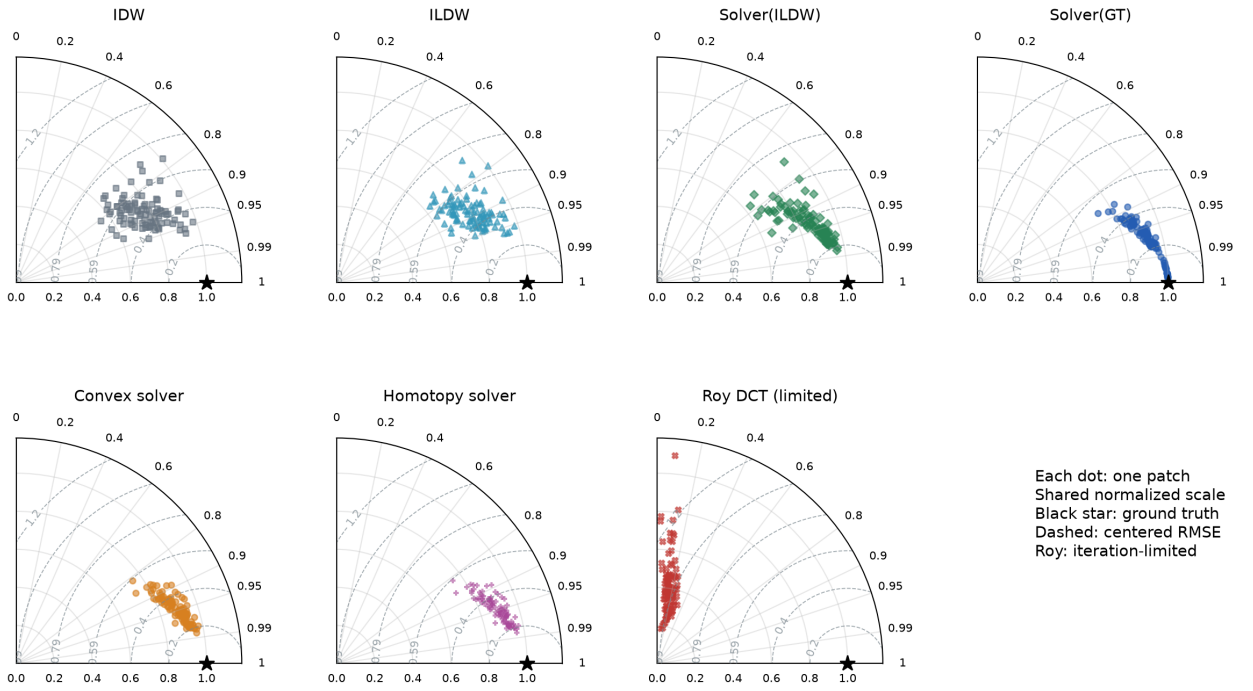


Figure 8: Individual normalized Taylor points: 100 saved patches for each method, on a common scale. Roy DCT points are from nonconverged, iteration-limited nonlinear runs.

7. Case Study Figures and Objective Evolution

The following two case studies retain the original selected methods and figures; they do not include the Roy adaptation. In this section, we present two case studies, in which we compare IDW, SOLVER(GT), and CONVEX SOLVER. In the first case study, SOLVER(GT) almost equals the ground-truth while CONVEX SOLVER slightly improves over IDW.

To quantify how often these two qualitative patterns occur, we use the simple ratio

$$q(P) \triangleq \frac{\text{RMSE}_{P,\text{CONVEX SOLVER}}}{\text{RMSE}_{P,\text{SOLVER(GT)}}}.$$

We assign patch P to Case 1 (the Case Study I pattern) when $q(P) \geq 2$ and to Case 2 (the Case Study II pattern) when $q(P) < 2$. Of the 100 benchmark patches, 24 are in Case 1 and 76 are in Case 2. The smallest Case 1 ratio is 2.6310, whereas the largest Case 2 ratio is 1.9767; thus no observed patch lies immediately at the threshold.

For this patch, $J(\text{SOLVER(GT)}) = 1.485$ and $\text{RMSE}(\text{SOLVER(GT)}) = 0.102$, while $J(\text{CONVEX SOLVER}) = 0.265$ and $\text{RMSE}(\text{CONVEX SOLVER}) = 0.702$. This example shows that a lower value of the objective does not necessarily imply a more accurate reconstruction of the ground-truth rainfall field. For Case Study I, Figure 11 shows the reconstructed rainfall fields and relative errors, and Figure 12 shows the evolution of the weighted objective over L-BFGS-B iterations.

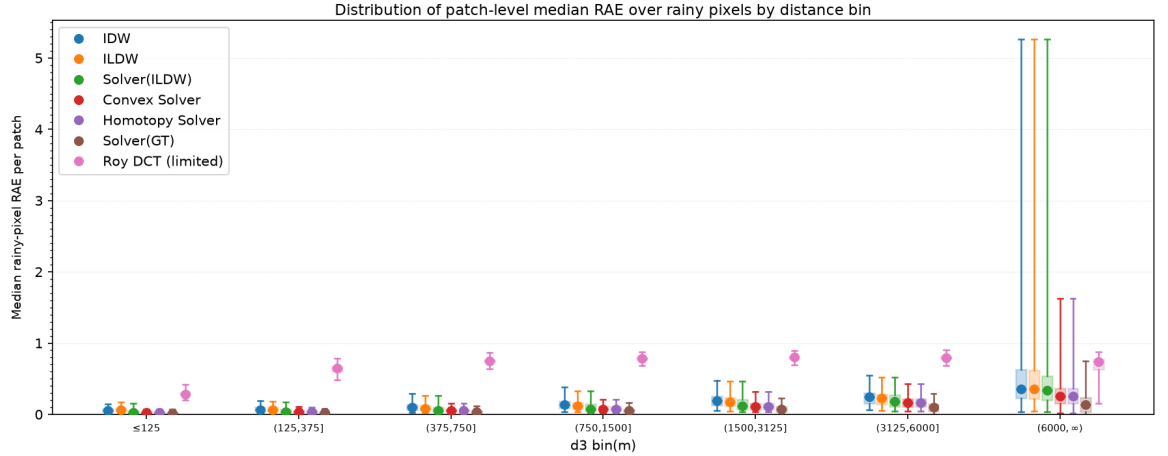


Figure 9: Rainy-pixel d_3 -profile box-and-whisker summary. Each patch contributes one median rainy-pixel relative absolute error value per d_3 bin. Across patches, each box spans the 25th to 75th percentiles, and the whiskers show the minimum and maximum values. The corresponding rainy-pixel counts per bin are reported in Table 4.

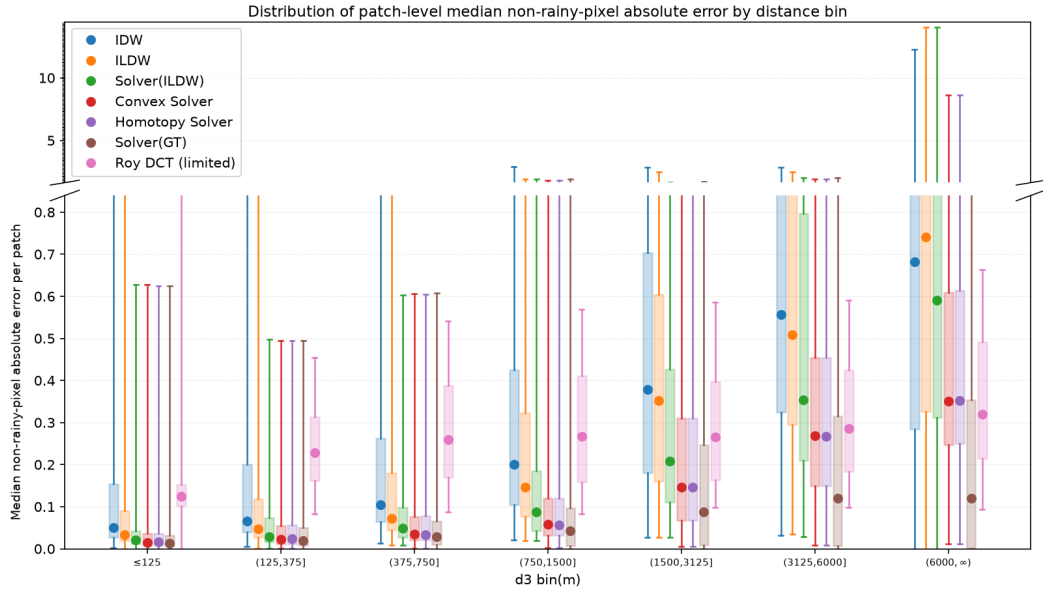


Figure 10: Non-rainy-pixel d_3 -profile box-and-whisker summary. Each patch contributes one median non-rainy-pixel absolute error value per d_3 bin. Across patches, each box spans the 25th to 75th percentiles, and the whiskers show the minimum and maximum values. The corresponding non-rainy-pixel counts per bin are reported in Table 4.

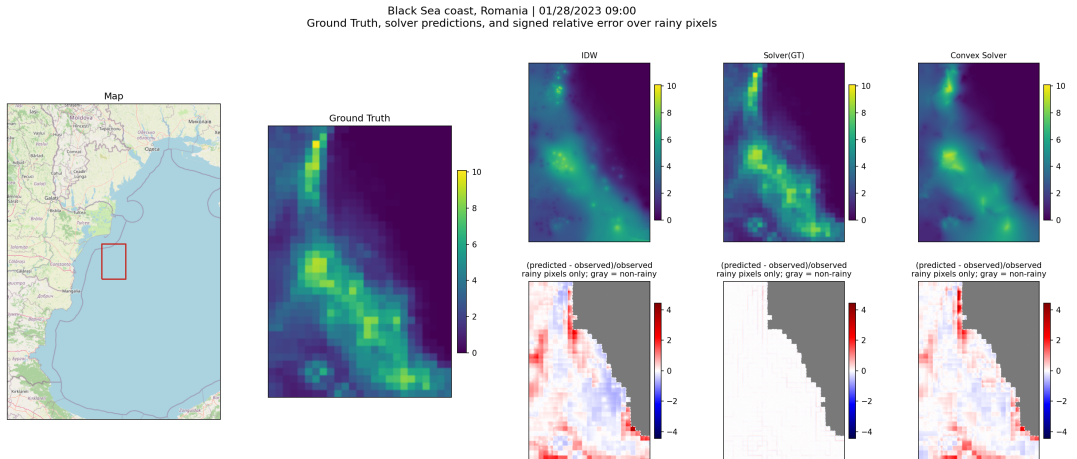


Figure 11: Case Study I: SOLVER(GT) is visually close to the ground truth. The three upper-right panels show the rainfall fields reconstructed by IDW, SOLVER(GT), and CONVEX SOLVER. The corresponding lower-right panels show the relative error of each reconstruction.

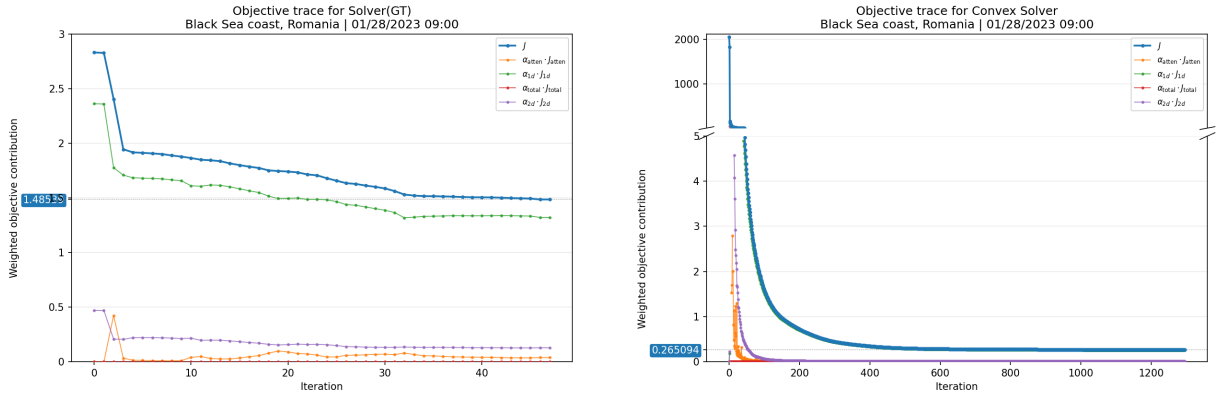


Figure 12: Evolution of the weighted objective over L-BFGS-B iterations for Case Study I. Left: SOLVER(GT), initialized from the ground-truth field, converges to final $J = 1.485$. Right: CONVEX SOLVER reaches a lower final weighted objective, $J = 0.265$, but its reconstruction is farther from the ground truth by RMSE.

In the second case study, SOLVER(GT) and CONVEX SOLVER are visually similar in the reconstruction panels. For this patch, $J(\text{SOLVER}(\text{GT})) = 0.289$ and $\text{RMSE}(\text{SOLVER}(\text{GT})) = 0.552$, while $J(\text{CONVEX SOLVER}) = 0.290$ and $\text{RMSE}(\text{CONVEX SOLVER}) = 0.565$. This example shows a case in which the two solutions have almost the same objective value and almost the same reconstruction error. For Case Study II, Figure 13 shows the reconstructed rainfall fields and relative errors, and Figure 14 shows the evolution of the weighted objective over L-BFGS-B iterations.

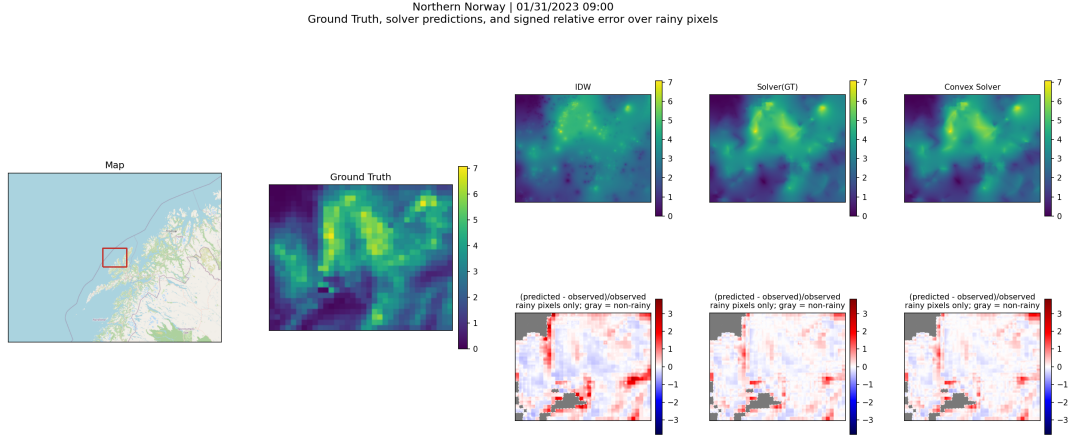


Figure 13: Case Study II: SOLVER(GT) and CONVEX SOLVER produce visually similar reconstructions. The three upper-right panels show the rainfall fields reconstructed by IDW, SOLVER(GT), and CONVEX SOLVER. The corresponding lower-right panels show the relative error of each reconstruction.

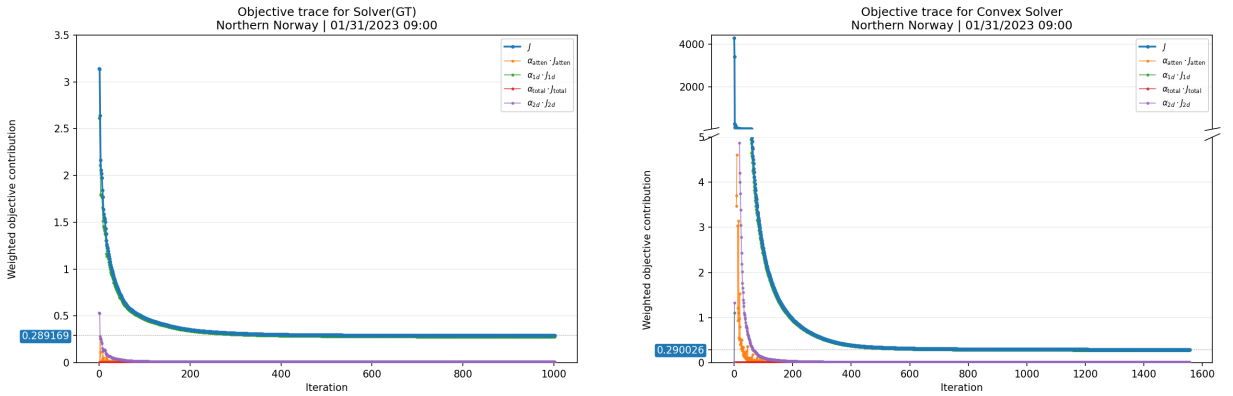


Figure 14: Evolution of the weighted objective over L-BFGS-B iterations for Case Study II. Left: SOLVER(GT), initialized from the ground-truth field, converges to final $J = 0.289$. Right: CONVEX SOLVER reaches a very similar final weighted objective, $J = 0.290$, and a similar RMSE.

8. Discussion

The added Roy DCT comparison is limited by incomplete nonlinear convergence in all 100 runs. Its low attenuation residuals and poor map metrics motivate separate convergence and hyperparameter studies; they do not establish a negative conclusion about the original linear method of Roy et al.

The benchmark isolates the geometric inversion problem and thereby provides a clear setting for method development. At the same time, several sources of uncertainty remain outside the scope of the current construction, including wet-antenna effects, baseline estimation, quantization, mixed-phase precipitation, and temporal dynamics. These factors matter in operational deployments and should be incorporated in later stages once the benchmarked inverse methods are well understood.

The proposed framework also offers interpretability advantages over purely data-driven alternatives. Because the objective function makes the tradeoff between attenuation consistency, spatial smoothness, and rainfall shrinkage explicit, the resulting reconstructions can be inspected and explained in terms of physically meaningful quantities.

The resulting optimization problem is nonconvex due to the nonlinear attenuation model. We therefore compute a locally optimal solution using L-BFGS-B, initialized from the ILDW estimate. In practice, the combination of strong data constraints and regularization yields stable and consistent reconstructions across instances.

The results highlight the importance of the initialization for the nonconvex optimization procedure. In many instances, the ILDW solution lies close to a stationary point of the objective, and the optimizer produces only marginal changes. This indicates that ILDW already achieves a strong balance between attenuation consistency and spatial regularity. Consequently, the role of the nonconvex optimization is primarily to refine the estimate in instances where these objectives are in stronger conflict. In contrast, IDW provides a weaker initialization due to its simplified geometric representation of links, leading to larger discrepancies with the attenuation model.

9. Conclusion

This journal draft integrates a benchmark-oriented view of rainfall reconstruction from commercial microwave links. Starting from hourly EURADCLIM maps, we extract rain events, simulate CML attenuations through a forward model, and evaluate inverse reconstruction methods against known ground truth. The resulting setup reframes rainfall reconstruction from opportunistic sensing networks as a well-posed inverse problem with an explicit and interpretable objective.

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A. Illustrative Examples

This appendix presents two minimal examples that isolate important properties of the inverse problem. First, for two parallel links on a 2×3 field, we show that midpoint IDW underestimates the rain rate at every pixel crossed by the second link except at the pixel where the link’s midpoint is located, whereas ILDW recovers the field exactly. Second, we give a concrete counterexample showing that J_{atten} is nonconvex.

A.1. IDW Underestimates Rain Along a Link

Consider the following ground-truth field

$$R^{\text{GT}} = \begin{bmatrix} 4 & 4 & 4 \\ 5 & 5 & 5 \end{bmatrix} \text{ mm h}^{-1}.$$

Let two equal-length horizontal links ℓ_1 and ℓ_2 cross the top and bottom rows, respectively. Their rain-rate equivalents are $R_{\ell_1} = 4$ and $R_{\ell_2} = 5 \text{ mm h}^{-1}$; we focus on the three pixels crossed by ℓ_2 . The link midpoints lie in the middle column, and the search radius includes both links.

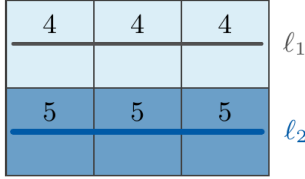
For either off-center pixel crossed by ℓ_2 , at row-column positions (2, 1) and (2, 3), the distances to the midpoints of ℓ_2 and ℓ_1 are h and $\sqrt{2}h$, where $h = 0.125 \text{ km}$ is the pixel side length. The normalized inverse-square weights in Algorithm 1 therefore give

$$\hat{R}_{2,1}^{\text{IDW}} = \hat{R}_{2,3}^{\text{IDW}} = \frac{5h^{-2} + 4(\sqrt{2}h)^{-2}}{h^{-2} + (\sqrt{2}h)^{-2}} = \frac{14}{3} \approx 4.67 \text{ mm h}^{-1}.$$

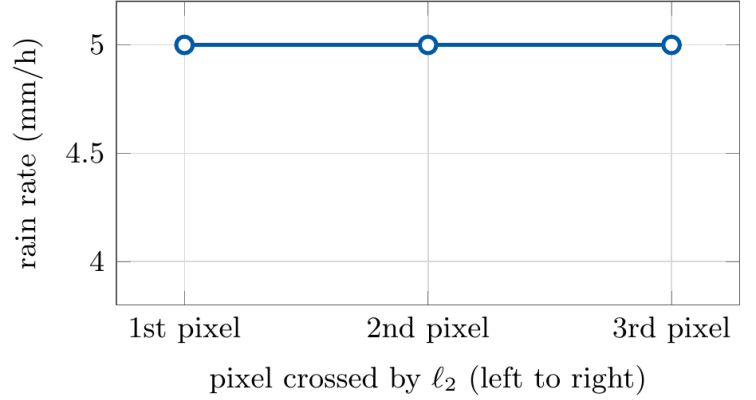
The middle pixel contains the midpoint of ℓ_2 , so $\hat{R}_{2,2}^{\text{IDW}} = 5 \text{ mm h}^{-1}$. Repeating the calculation over the grid yields

$$\hat{R}^{\text{IDW}} = \begin{bmatrix} 13/3 & 4 & 13/3 \\ 14/3 & 5 & 14/3 \end{bmatrix} \approx \begin{bmatrix} 4.33 & 4 & 4.33 \\ 4.67 & 5 & 4.67 \end{bmatrix}.$$

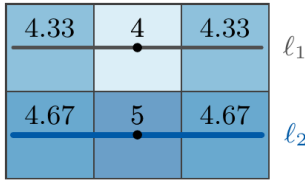
By contrast, every pixel in a row is crossed by that row's link, so the crossing-pixel rule in Algorithm 2 gives $\hat{R}^{\text{ILDW}} = R^{\text{GT}}$. Thus the profiles along ℓ_2 are (14/3, 5, 14/3) for IDW and (5, 5, 5) for ILDW. These calculations apply the paper's IDW and ILDW procedures exactly as stated in Algorithms 1 and 2.



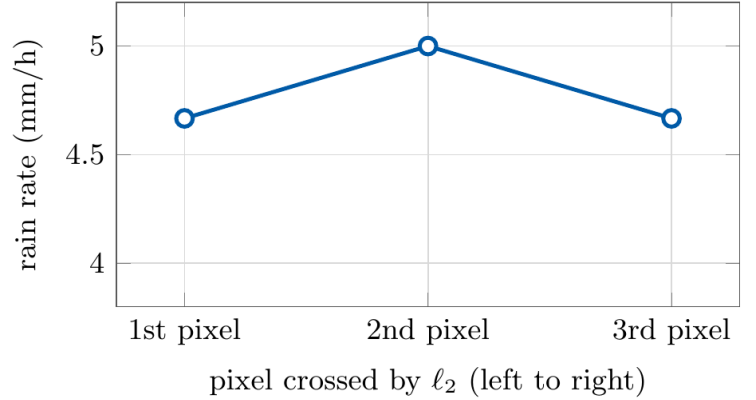
(a) Ground truth and link geometry



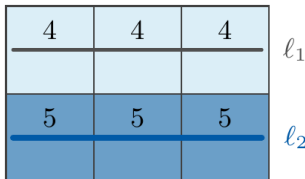
(b) Ground-truth profile along ℓ_2



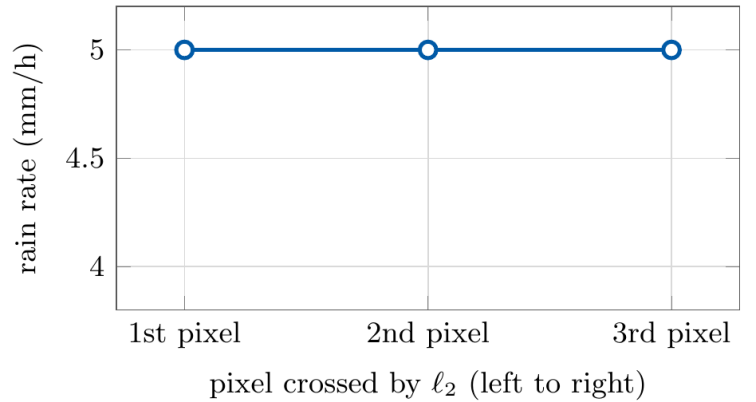
(c) IDW reconstruction



(d) IDW profile along ℓ_2



(e) ILDW reconstruction



(f) ILDW profile along ℓ_2

Figure 15: Geometric difference between midpoint IDW and ILDW. Away from the middle column, IDW mixes the 5 mm h⁻¹ observation from ℓ_2 with the neighboring 4 mm h⁻¹ observation. ILDW associates each link-level observation with every pixel crossed by its segment and therefore exactly recovers this row-wise constant field.

The second link is essential for this normalized-IDW example: if ℓ_2 were the only link, its single normalized weight would cancel, and IDW would return $R_{\ell_2} = 5$ at every in-radius pixel. The IDW underestimation away from the midpoint and the exact ILDW recovery are illustrated by the six panels in Figure 15.

A.2. A 3×3 Counterexample to Convexity

Consider one horizontal link ℓ crossing the three middle-row pixels of a 3×3 grid. The crossed length is 0.125 km in each pixel, so $|\ell| = 0.375$ km. Let the link use horizontal polarization at 38.003 GHz, for which the ITU-R coefficients used here are $\kappa = 0.400171594750$ and $\alpha = 0.881535333268$. For this one-link instance,

$$\hat{A}(R) = \sum_{p \in \ell} d_p \kappa R(p)^\alpha, \quad J_{\text{atten}}(R) = \frac{(\hat{A}(R) - A_{\text{obs}})^2}{|\ell|}.$$

Take the two strictly positive fields

$$R^{(L)} = \begin{bmatrix} 1 & 1 & 1 \\ 20 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}, \quad R^{(R)} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 20 \\ 1 & 1 & 1 \end{bmatrix} \text{ mm h}^{-1}.$$

They are left-right reflections across the grid's vertical centerline. Either field could be the ground truth: the link crosses the reflected pixels over equal lengths, so both fields produce

$$\hat{A}(R^{(L)}) = \hat{A}(R^{(R)}) = A_{\text{obs}} = 0.801595407171 \text{ dB},$$

and hence $J_{\text{atten}}(R^{(L)}) = J_{\text{atten}}(R^{(R)}) = 0$. Their arithmetic mean is

$$R^{(M)} = \frac{R^{(L)} + R^{(R)}}{2} = \begin{bmatrix} 1 & 1 & 1 \\ 10.5 & 1 & 10.5 \\ 1 & 1 & 1 \end{bmatrix}.$$

For $g(r) = r^\alpha$, we have $g''(r) = \alpha(\alpha-1)r^{\alpha-2} < 0$ because $0 < \alpha < 1$; hence g is strictly concave and $2(10.5)^\alpha > 20^\alpha + 1$. Consequently,

$$\hat{A}(R^{(M)}) = 0.845082934781 \text{ dB} > A_{\text{obs}}, \quad J_{\text{atten}}(R^{(M)}) = 0.005043106820 > 0.$$

Therefore,

$$J_{\text{atten}}\left(\frac{R^{(L)} + R^{(R)}}{2}\right) > \frac{J_{\text{atten}}(R^{(L)}) + J_{\text{atten}}(R^{(R)})}{2} = 0,$$

which violates the defining inequality for a convex function. Thus the physical attenuation term is nonconvex even in this one-link 3×3 instance.